BJT: Operation, Load Line and Switch



Basic Electrical and Electronics Engineering

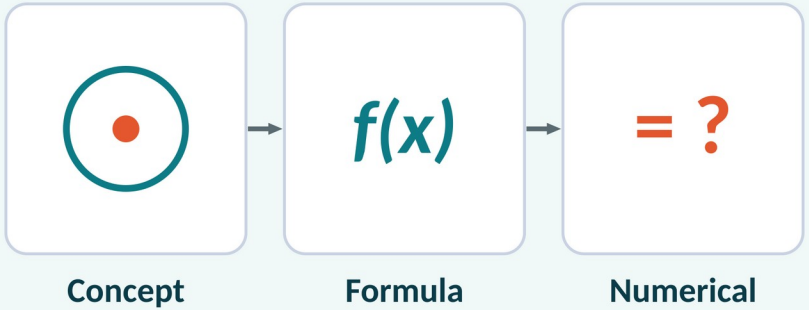
Unit 4: Transistors

Topic: BJT construction and operation, characteristics, DC load line, BJT as a switch

Prof. Mitesh Astik

Learning Outcomes

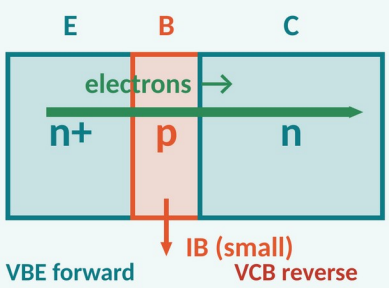
- Describe BJT construction and operation.
- Use IC = βIB and IE = IB + IC.
- Read output characteristics and the DC load line.
- Design a BJT switch.



Linked to CO4
Describe the construction and operation of BJT to operate as a switch.

Recap: What You Already Know

- p-n junction: forward and reverse bias.
- Majority and minority carriers.
- Forward bias lowers the barrier.
- Kirchhoff's current law at a node.



1 EB junction forward biased: emitter inj

2 Base is thin: few recombine → small IB

3 Most electrons are swept to the collect

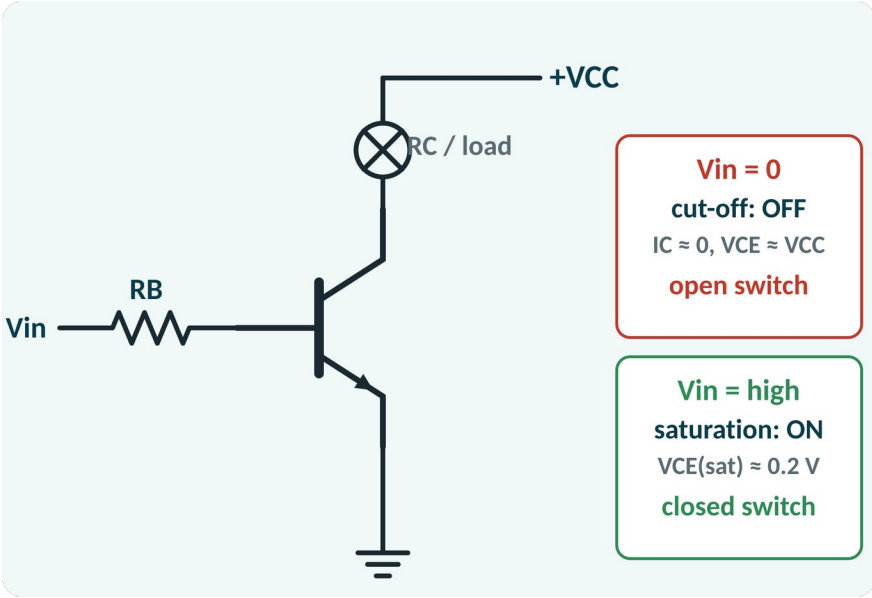
4 CB junction reverse biased: collects the

IE = IB + IC

KCL: IE = IB + IC; Vγ ≈ 0.7 V for a forward-biased silicon junction

Motivation: One Device, Two Jobs

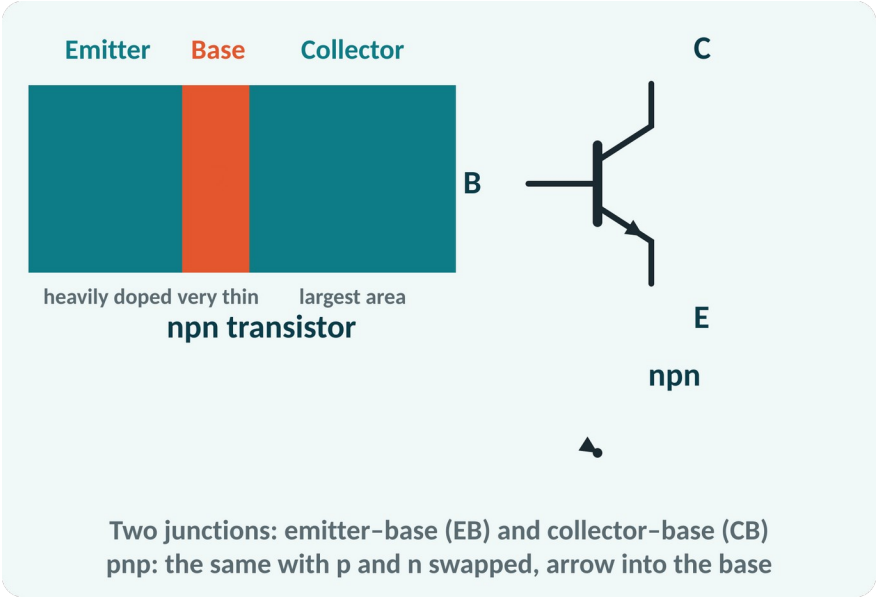
- Amplifier: small signal controls big current.
- Switch: turns lamps, relays and motors on and off.
- Processors use millions of switches.
- Interface between logic and power.



A small base current controls a much larger collector current

BJT Construction

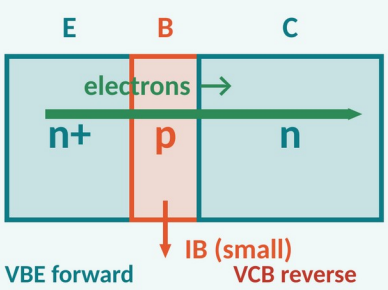
- Three layers: emitter, base, collector.
- npn: n–p–n sandwich; pnp: p–n–p.
- Emitter: heavily doped, emits carriers.
- Base: very thin, lightly doped.
- Collector: largest, collects carriers.



Arrow on the emitter shows conventional current direction (out for npn, in for pnp)

Operation of an npn Transistor

- Active mode: EB forward, CB reverse.
- Emitter injects electrons into the base.
- Base is thin: only about 1 % recombine → IB.
- Reverse CB field sweeps the rest to the collector.
- Small IB controls large IC.



1 EB junction forward biased: emitter inj

2 Base is thin: few recombine → small IB

3 Most electrons are swept to the collect

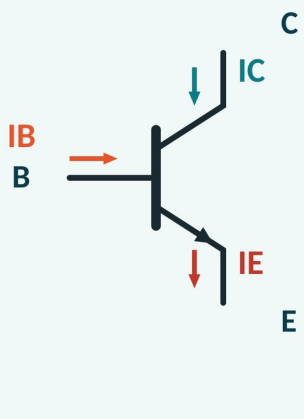
4 CB junction reverse biased: collects the

IE = IB + IC

IE = IB + IC; IC ≈ IE (α ≈ 0.99)

Current Relations

- β (hFE): current gain, typically 50–300.
- α: fraction of IE reaching the collector.
- IC = β·IB in the active region.
- Emitter current is the largest.



$I_C = \beta \cdot I_B$

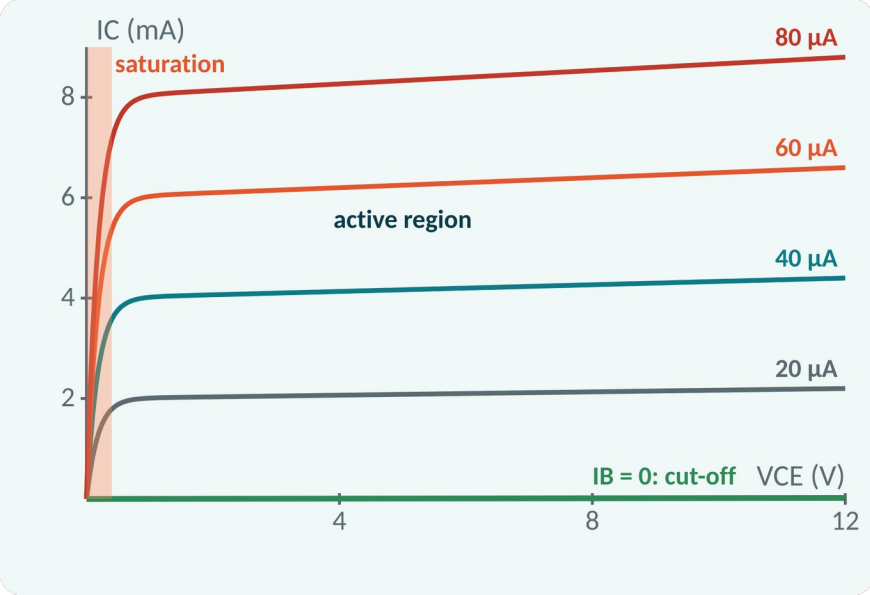
$I_E = I_B + I_C = (\beta + 1) \cdot I_B$

$\alpha = I_C / I_E = \beta / (\beta + 1) \approx 1$

$I_C = \beta \cdot I_B$; $I_E = I_B + I_C = (\beta + 1) \cdot I_B$; $\alpha = \beta / (\beta + 1)$
β dimensionless, currents in A or mA

Common-Emitter Output Characteristics

- IC against VCE for different base currents.
- Active region: curves nearly flat, IC ≈ β·IB.
- Saturation: small VCE, IC rises steeply.
- Cut-off: IB = 0, IC ≈ 0.
- Spacing of curves reflects β.



Active: IC ≈ β·IB; saturation: VCE(sat) ≈ 0.2 V; cut-off: IC ≈ ICEO (tiny)

Regions of Operation

- Cut-off: both junctions reverse biased.
- Active: EB forward, CB reverse.
- Saturation: both junctions forward biased.
- Amplifier works in the active region.
- Switch works between cut-off and saturation.

Region	EB junction	CB junction	Use
Cut-off	reverse	reverse	switch OFF
Active	forward	reverse	amplifier
Saturation	forward	forward	switch ON

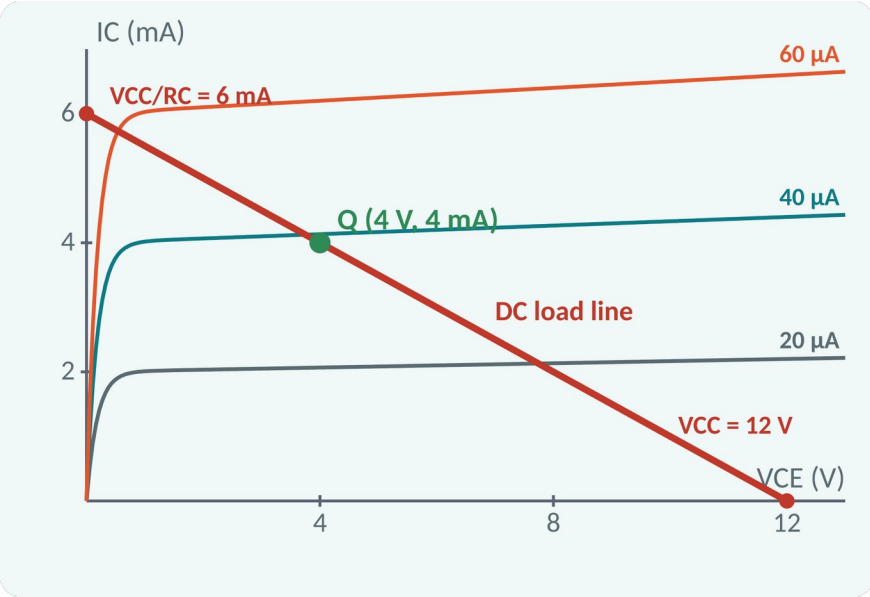
BJT is a current-controlled device

$I_C \approx \beta \cdot I_B$ only in the active region

Cut-off: $I_B = 0$, switch OFF; Saturation: $I_C = I_{C(sat)}$, switch ON

DC Load Line

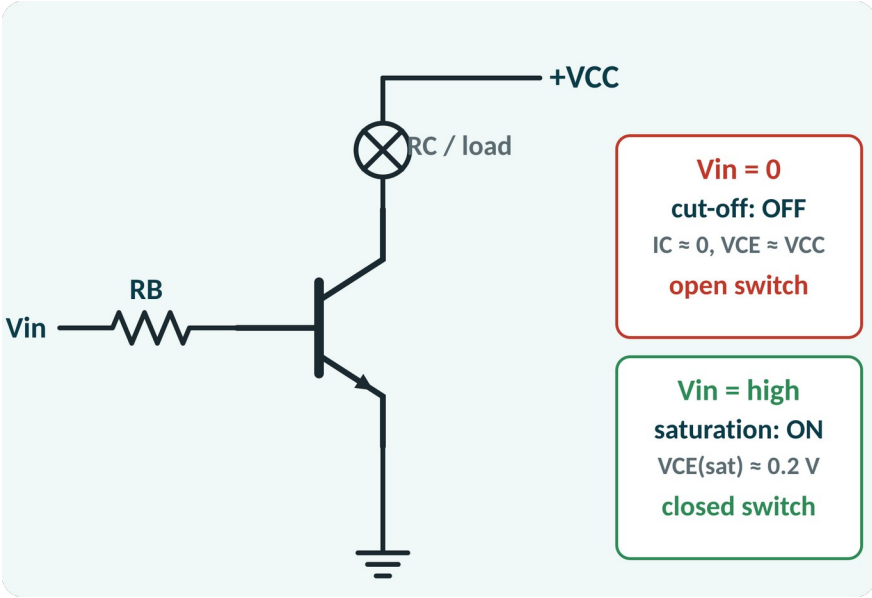
- Collector circuit: VCC = IC·RC + VCE.
- Straight line on the output characteristics.
- End points: (VCE = VCC, IC = 0) and (0, VCC/RC).
- Operating (Q) point lies on the load line.
- Q is set by the base current.



IC = (VCC − VCE)/RC; example VCC = 12 V, RC = 2 kΩ: (12 V, 0) and (0, 6 mA)
Q: IB = 40 µA → IC = 4 mA, VCE = 4 V

BJT as a Switch

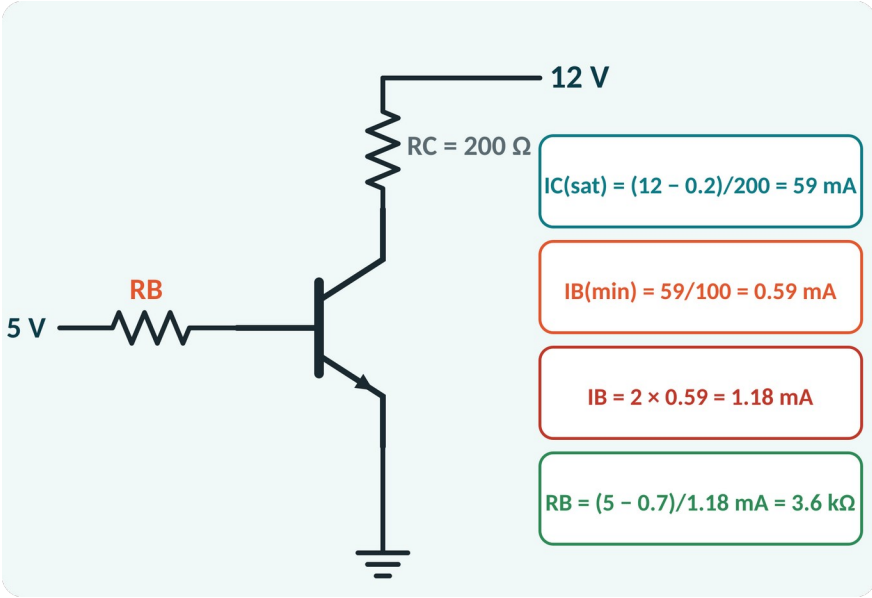
- Input low: cut-off, transistor OFF.
- Input high: saturation, transistor ON.
- Load gets full VCC when ON, nothing when OFF.
- VCE(sat) ≈ 0.2 V: nearly an ideal closed switch.
- Used for LEDs, relays, motors and logic gates.



OFF: IC ≈ 0, VCE ≈ VCC; ON: IC = (VCC − VCE(sat))/RC, VCE ≈ 0.2 V

Designing a BJT Switch

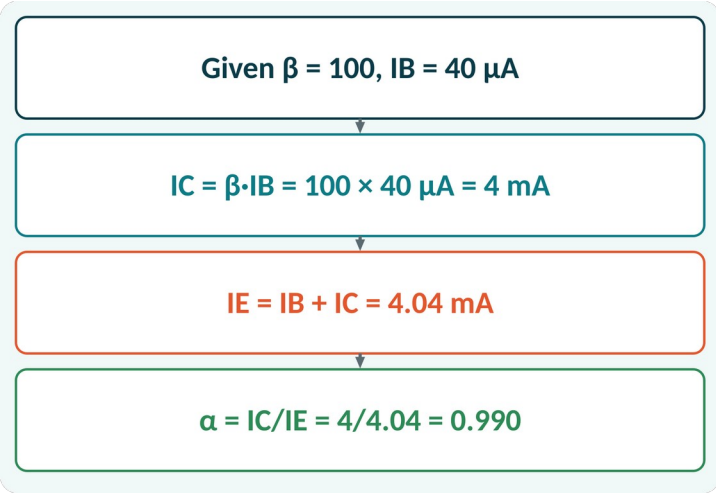
- Find IC(sat) from the load.
- Required IB(min) = IC(sat)/βmin.
- Overdrive: use 2–5 × IB(min).
- Calculate RB from the input voltage.
- Choose the nearest standard value.



IC(sat) = (VCC – VCE(sat))/RC; IB ≥ IC(sat)/β; RB = (Vin – VBE)/IB

Solved Example 1: BJT Currents

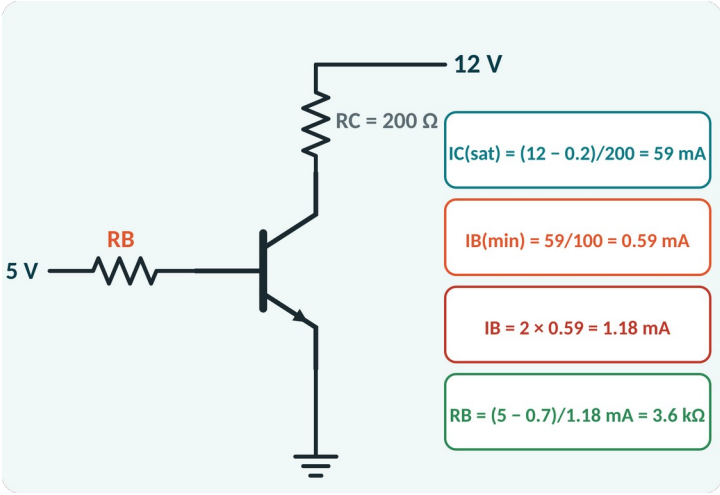
- A transistor with β = 100 has IB = 40 μA in the active region. Find IC, IE and α.
- 1. IC = β·IB = 100 × 40 μA = 4 mA.
- 2. IE = IB + IC = 0.04 + 4 = 4.04 mA.
- 3. α = IC/IE = 4/4.04 = 0.990.
- 4. Check: α = β/(β + 1) = 100/101 = 0.990.



Answer: IC = 4 mA, IE = 4.04 mA, α = 0.990

Solved Example 2: Switch Design

- Drive a 200 Ω load from 12 V using a 5 V logic signal. βmin = 100, VBE = 0.7 V, VCE(sat) = 0.2 V. Find RB.
- 1. IC(sat) = (12 − 0.2)/200 = 59 mA.
- 2. IB(min) = 59 mA/100 = 0.59 mA.
- 3. Overdrive × 2: IB = 1.18 mA.
- 4. RB = (5 − 0.7)/1.18 mA = 3.64 kΩ.
- 5. Choose RB = 3.3 kΩ: IB = 1.30 mA, which saturates the transistor.



Answer: RB ≈ 3.3 kΩ (standard), IC(sat) = 59 mA

Common Mistakes and Exam Tips

Common mistakes

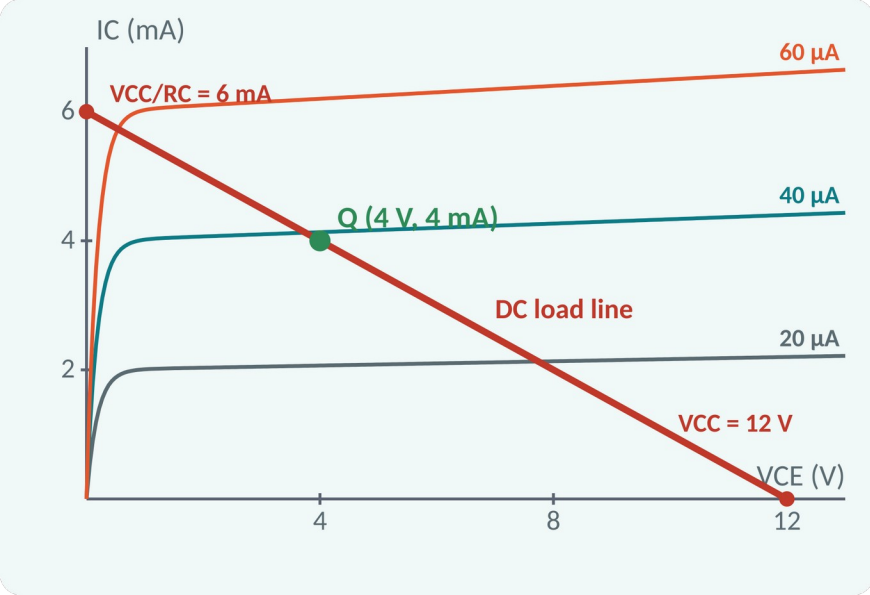
- Using IC = β·IB in saturation.
- Forgetting VBE = 0.7 V in the RB calculation.
- Designing the switch with typical β instead of minimum.
- Connecting a pnp with npn polarity.
- No base resistor (damages the junction).

Exam tips

- Check the region before using IC = βIB.
- Overdrive the base for a switch.
- Use the minimum β from the data sheet.
- Check power rating of the transistor.

Summary

- BJT: emitter, base, collector; two junctions.
- Active: EB forward, CB reverse.
- IC = β·IB; IE = IB + IC.
- DC load line joins (VCC, 0) and (0, VCC/RC).
- Switch: cut-off = OFF, saturation = ON.



RB = (Vin − VBE)/IB, IB ≥ IC(sat)/β

Quick Quiz

Q1. In active mode the junctions are:
(a) EB forward, CB reverse (b) both forward (c) both reverse (d) EB reverse, CB forward

Ans: (a)

Q2. β = 100, IB = 20 µA gives IC =
(a) 0.2 mA (b) 2 mA (c) 20 mA (d) 2 µA

Ans: (b)

Q3. BJT switch is ON in:
(a) cut-off (b) active (c) saturation (d) breakdown

Ans: (c)

Q4. Load-line saturation point for VCC = 10 V, RC = 1 kΩ:
(a) 10 mA (b) 1 mA (c) 100 mA (d) 10 V

Ans: (a)

Q5. The base of a BJT is:
(a) thick, heavily doped (b) thin, lightly doped (c) the largest region (d) metal

Ans: (b)

References

1. **Boylestad Robert L. and Nashelsky Louis, Electronic Devices and Circuit Theory**
   11th Ed., PHI, 2016 (Units 3, 4, 5 textbook)

2. **H. Cotton, Electrical Technology**
   7th Ed., CBS Publishers, 2005 (reference)

Next lecture: MOSFET, common-source amplifier and MOS inverter